Faraday rotation with the LuSEE-Night four-port correlator: zoom-bin spectroscopy of the polarized sky

August 17, 2026

Abstract

We redo the LuSEE-Night Faraday-rotation forecast on the full four-port instrument model, producing all sixteen correlation products of the N, E, S, W monopoles, and propagate them through the actual spectrometer channelization, including the 64 zoom sub-bins of the 25 kHz parent channels. A single fully polarized source behind a $\phi_{\rm FD} = 250$ rad m^{-2} screen produces polarization oscillations with a ∼1.9 kHz period at 30 MHz: erased to better than 10^{-3} in the parent channels, but recovered at ∼79–86% amplitude by the zoom bins (with a zenith-calibrated four-port polarimeter whose instrumental leakage is nulled exactly). [Steps 2–4: realistic sky, 10/50 MHz bands, delay-space signature – to be filled in.]

1 Simulation setup

The instrument model is the luseepy four-port response (BGL_v16 v3 artifact): coupled 2× monopole fields $H_p^{\theta,\phi}$ for the ports $p \in \{\mathrm{N}, \mathrm{E}, \mathrm{S}, \mathrm{W}\}$, antenna impedance matrix Z_A and the JFET receiver loading $M = Z_L(Z_A + Z_L)^{-1}$. For a sky coherency written in the tangent basis (e_θ, e_ϕ) of the topocentric frame,

$$B_E = \frac{k_B \eta_0}{\lambda^2} \begin{pmatrix} I+Q & U-jV \\ U+jV & I-Q \end{pmatrix}, \tag{1}$$

the open-circuit pair correlations are beam-weighted integrals of the pair-Stokes kernels $P_{ab}^{I,Q,U,V}$, and the loaded covariance is $C = MKM^\dagger$. The sixteen telemetered products are the four autos and the real/imaginary parts of the six crosses.

The analysis band is a ± 0.1 MHz window around a band center $\nu_0 \in \{30, 10, 50\}$ MHz, sampled by a fine grid of 16384 frequencies spaced at 25 kHz/2048 = 12.2 Hz. The beam is deliberately frozen at the native ν_0 channel (the fixed-beam approximation), so that any spectral structure is purely the Faraday phase

$$(Q + iU)(\nu) = (Q + iU)\, e^{+2i\phi_{\rm FD}\lambda^2(\nu)}. \tag{2}$$

Time is sampled at 1024 points over exactly one lunar sidereal day (27.32 d), making the time axis periodic and FFT-ready. The observatory is at the LuSEE-Night site (−23.814°, 182.258°).

The fine waterfalls (time × frequency × 16 products) are integrated against the measured spectrometer response into (i) the three 25 kHz parent channels covered by the window, (ii) their 3 × 64 zoom sub-bins (390.625 Hz spacing, FFT ordering), and (iii) an idealized set of Gaussian zoom bins of matching FWHM at the nominal sub-bin centers.

The pixel-space engine was validated against the luseepy harmonic engines: point-source covariances agree with `FullStokesCalibratorSimulator` to 9×10^{-16} relative, and the diffuse-sky transport (frame rotation, polarization spin, COSMO/IAU conventions, absolute normalization) against `FullStokesCroSimulator`.

2 Step 0: an unpolarized source and the calibrated polarimeter

We first characterize the instrument itself on a transiting *unpolarized* point source (1 K sr), placed so that it culminates 0.6° from zenith; it is above the horizon 56% of the sidereal day. With the frozen beam and no polarization the response is exactly frequency-flat — parent bins, zoom bins and fine channels are identical — so any recovered pseudo-polarization is pure instrumental leakage. Figure 1 shows this leakage as a function of source altitude for the raw $X = \mathrm{E}-\mathrm{W}$, $Y = \mathrm{N}-\mathrm{S}$ polarimeter: an ideal crossed-dipole polarimeter would give zero leakage at zenith, but the as-built four-port beams retain $p \simeq 0.13$ even at transit, rising to $\simeq 0.94$ at 30°–35° altitude.

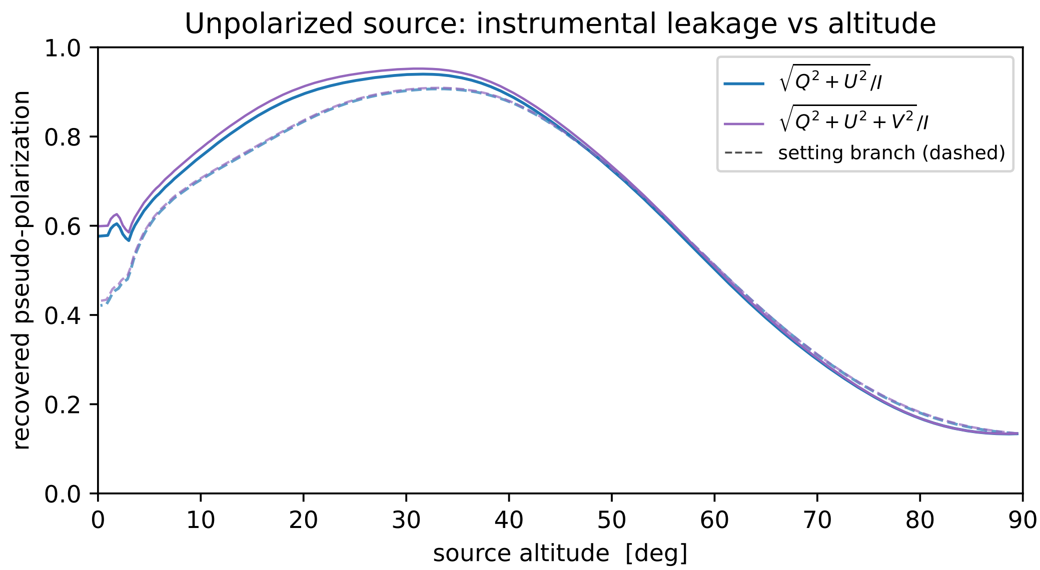

Figure 1: Unpolarized transiting source, *raw* $X = \mathrm{E}-\mathrm{W}$, $Y = \mathrm{N}-\mathrm{S}$ polarimeter: the recovered pseudo-polarization is pure instrumental leakage (frequency-flat, so all channelizations coincide). Solid: rising branch, dashed: setting branch.

Zenith-calibrated polarimeter. The zenith leakage reflects the asymmetry of the as-built beams: an unpolarized source at exact zenith produces autos $(NN, EE, SS, WW) \propto (1.26, 1.12, 1.25, 1.00)$ and the raw polarimeter reports $(q, u, v) = (-0.094, -0.096, +0.013)$. We calibrate the polarimeter on this zenith response in two stages. First, per-port gain weights $X = w_E E - w_W W$, $Y = w_N N - w_S S$ with $w_p \propto C_{pp}^{-1/2}$, i.e. $(w_N, w_E, w_S, w_W) = (0.952, 1.019, 0.956, 1.079)$ (plus a common X/Y rescaling), null the zenith pseudo-Q exactly but leave $u = -0.096$: that term enters through the inter-port cross-couplings ($\langle NE^* \rangle$ etc.) in $\langle YX^* \rangle$ and no real per-port gain can touch it. Second, we therefore let X and Y be *complex combinations of all four ports*: the zenith conditions $Q = U = V = 0$ for pure I are precisely the statement that x and y are orthogonal with equal norms in the metric of the zenith covariance C_0, so a symmetric (Löwdin) $G^{-1/2}$ orthonormalization of the gain-weighted pair achieves them exactly while staying as close as possible to the E–W / N–S dipoles. The result,

$$
\begin{aligned}
X &= 1.023\,E - 1.082\,W + (0.046 + 0.004i)(N - S), \\
Y &= 0.955\,N - 0.959\,S + (0.049 - 0.004i)\,E - (0.052 - 0.004i)\,W,
\end{aligned}
$$

leaves zenith leakage of $(q,u,v) \sim 10^{-16}$ — zero to machine precision. Figure 2 repeats the unpolarized track with these vectors: the transit leakage collapses from 0.134 to 7×10^{-4} (set only by the source culminating 0.6° off exact zenith). Away from zenith the leakage is essentially unchanged — the calibration enforces a single condition at one direction — so compact-source polarimetry remains leakage-limited at low altitude. All polarimeter results in the remainder of this report use the calibrated vectors; the raw-polarimeter figure versions (step1_* instead of step1w_* in `report/figures`) are retained for reference.

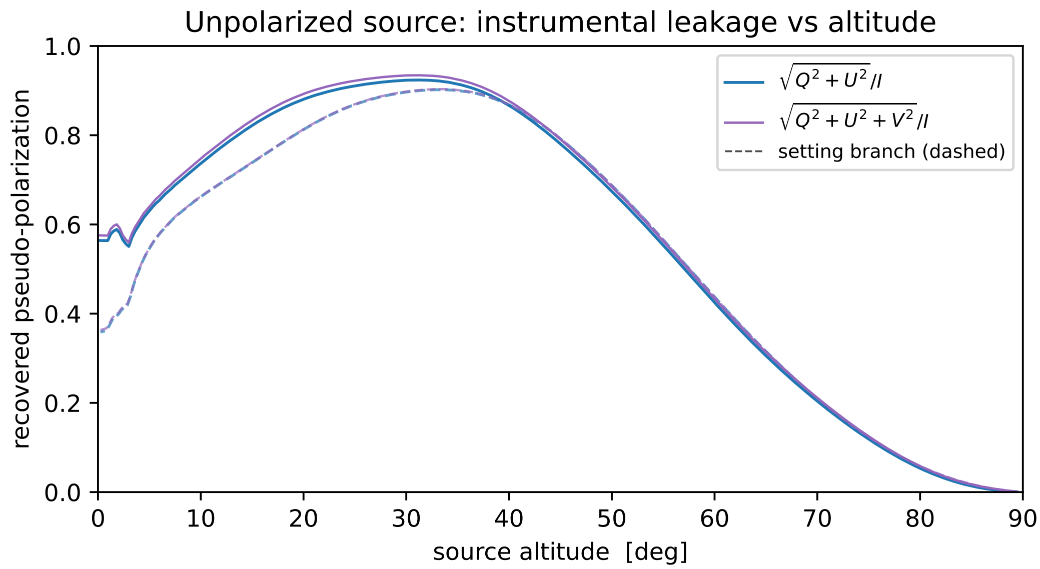

Figure 2: As Fig. 1, with the zenith-calibrated four-port polarimeter. The leakage now vanishes towards transit ($p \to 7 \times 10^{-4}$ at 0.6° from zenith), while remaining $O(1)$ at low altitudes where the single-point zenith calibration has no power.

3 Step 1: a single polarized source

The same transiting source, now fully polarized with its electric vector fixed along ecliptic north. The instrument-frame polarization angle is $\chi(t,\nu) = \psi(t) + \phi_{\rm FD}\lambda^2(\nu)$ with ψ the parallactic angle and $\phi_{\rm FD} = 250\,\mathrm{rad\,m^{-2}}$. All pseudo-Stokes below use the zenith-calibrated polarimeter of Sec. 2.

At 30 MHz the Faraday oscillation period is $\Delta\nu = \pi\nu_0^3/(2\phi_{\rm FD}c^2) \simeq 1.9\,\mathrm{kHz}$ (Fig. 3): about 13 full $Q \to U \to Q$ rotations fit inside one 25 kHz parent channel, while a zoom bin samples only ~ 0.2 of a rotation. Figure 4 zooms into $\pm 3\,\mathrm{kHz}$ of the same spectrum, where the individual oscillations and the amplitude transfer of the real and ideal zoom bins are visible. The recovered polarization fraction of every frequency bin at transit is collected in Fig. 5, and Fig. 6 follows it along the whole track as a function of source altitude. Finally, Fig. 7 shows the four correlator combinations of the calibrated polarimeter — $\langle|X|^2\rangle$, $\langle|Y|^2\rangle$, $\mathrm{Re}\,\langle XY^*\rangle$, $\mathrm{Im}\,\langle XY^*\rangle$ — as waterfalls over the track; the full set of sixteen raw N/E/S/W products is available as the step1_product_waterfalls figure in `report/figures`.

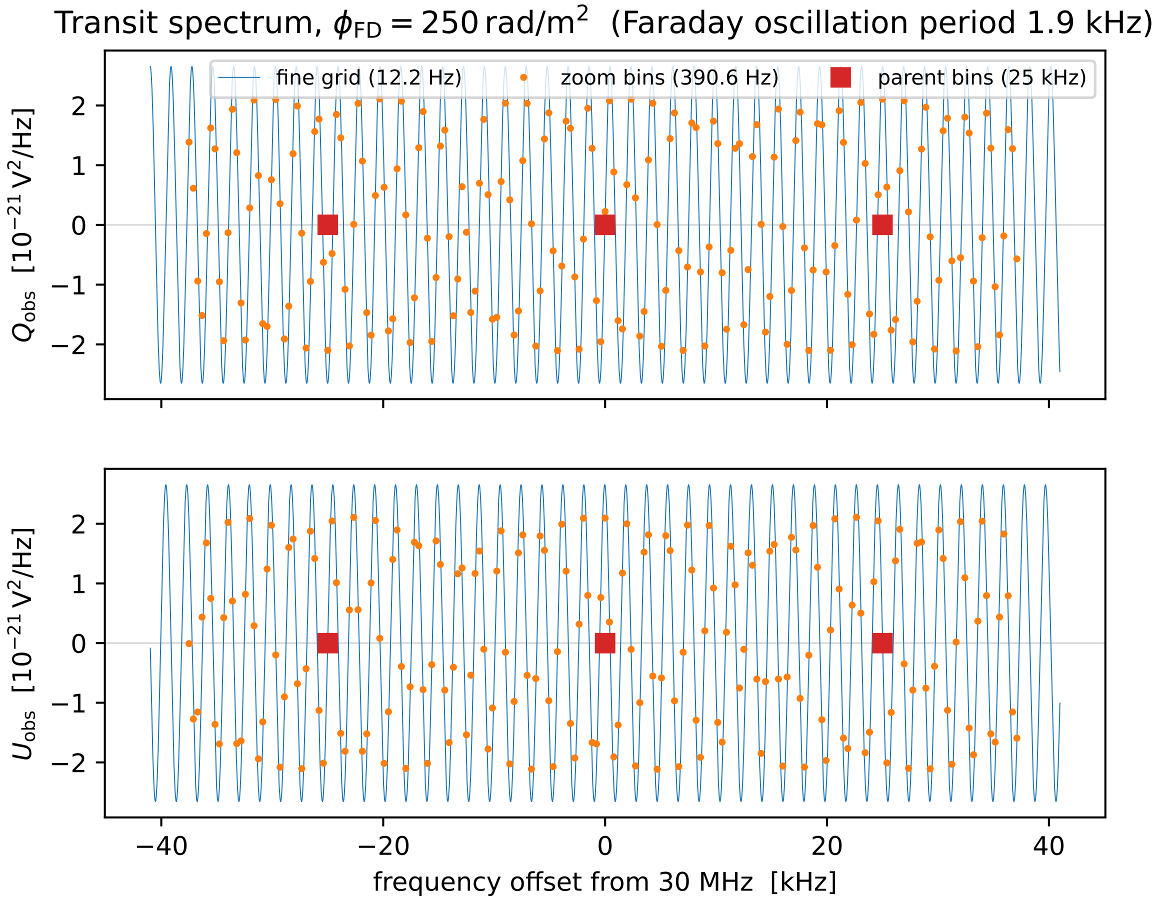

Figure 3: Observed pseudo-Stokes $Q_{\mathrm{obs}}, U_{\mathrm{obs}}$ (zenith-calibrated polarimeter) at transit as a function of frequency. The fine grid (blue) resolves the 1.9 kHz Faraday oscillation; the zoom bins (orange) trace it with a mild amplitude suppression; the parent 25 kHz bins (red) are almost completely depolarized.

4 Step 2: the real sky at 30 MHz

The sky model (Fig. 8) combines the destriped, desourced Haslam 408 MHz map scaled with $\beta = -2.55$ for I, the WMAP K-band (23 GHz) Q, U maps scaled with $\beta = -2.8$, and the Hutschenreuter & Enßlin (2020) mean Faraday-depth sky. All maps are used at their native $N_{\mathrm{side}} = 512$: the per-pixel Faraday phase does not commute with pixel-averaging, so the maps are never degraded; the pixel-space engine sums the full-resolution maps directly. Here and in everything that follows, the pseudo-Stokes are formed with the zenith-calibrated polarimeter of Sec. 2, using the weights of the respective band center.

5 Step 3: 10 and 50 MHz bands

The same analysis was repeated with the band centered at 10 and 50 MHz (no additional point source: a source is fully degenerate with a rotation-measure change). The channel-averaged behaviour mirrors 30 MHz — the smooth part of the beam-integrated polarization is spectrally re-

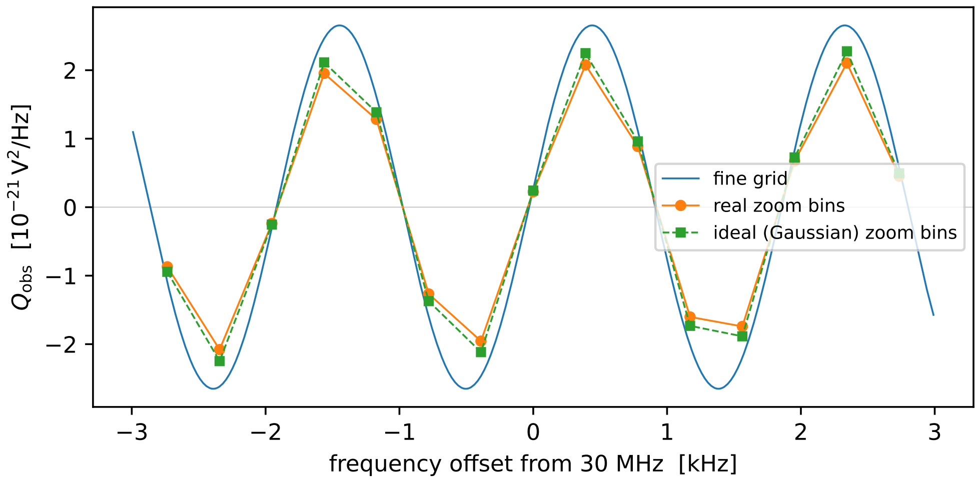

Figure 4: Zoom into ± 3 kHz around the band center. Real (measured response) and ideal (Gaussian) zoom bins track the fine spectrum with $\simeq 0.79$ and $\simeq 0.86$ of the amplitude respectively.

solved by every channelization — but the amplitude of the fast Faraday "fuzz" superimposed on it grows dramatically towards low frequency, as each pixel's oscillation rate scales as ν^{-3}.

6 The perfect-depolarization limit: an I-only sky

How much do the telemetered products actually care about sky polarization once the Faraday screen has scrambled it? To answer this we reran the real-sky simulation with $Q = U = 0$ — the limit of perfect Faraday depolarization at the band center. With the frozen beam and no polarization this sky is exactly frequency-flat, so its parent bins, zoom bins and fine channels are all identical, and the comparison against the full run isolates the polarization contribution to every product.

Two facts emerge. First, the polarization that *is* observed through the Faraday screen ($|P_{\rm obs}|/I_{\rm obs} \sim 0.15$ at 30 MHz, cf. Fig. 9) is almost entirely instrumental leakage of the unpolarized emission: the I-only run reproduces the 30 MHz parent-bin $(Q_{\rm obs}, U_{\rm obs})/I_{\rm obs} = (0.146, -0.032)$ of the full run to within 2×10^{-4}. (The zenith calibration of Sec. 2 does not reduce this diffuse-sky leakage — it enforces a single condition at one direction, while the whole visible hemisphere contributes here.) Without the Faraday screen the genuine sky polarization would contribute a median $|\Delta P_{\rm obs}|/I_{\rm obs} \simeq 6.2 \times 10^{-2}$ at 30 MHz; the beam-scale rotation-measure mixture suppresses this by a factor $\simeq 380$ in the parent bin.

Second, the residual genuine-polarization signal is channelization-dependent (Table 1 and Fig. 14). At 30 MHz the sky polarization moves the parent-bin products by a median 1.6×10^{-4} of $I_{\rm obs}$ (max 5.3×10^{-4}) and the central zoom bin by 4.5×10^{-4} (max 9.0×10^{-4}) — the finer bin averages away less of the rapidly rotating polarization and retains $\sim 2.7\times$ more of it. The contrast is strongest at 10 MHz (factor ~ 4 in $|\Delta P|$, factor ~ 5 in ΔI), where the Faraday oscillations are fastest, and nearly disappears at 50 MHz where even a 25 kHz bin resolves most of the structure.

For calibration and foreground modeling this is good news twice over: the total-power products are insensitive to sky polarization at the few$\times 10^{-4}$ level even in the zoom bins, while the polarization products themselves must be interpreted as leakage-dominated, with the genuine Faraday-surviving sky signal accessible only differentially (e.g. through the delay-space statistics of Sec. 7).

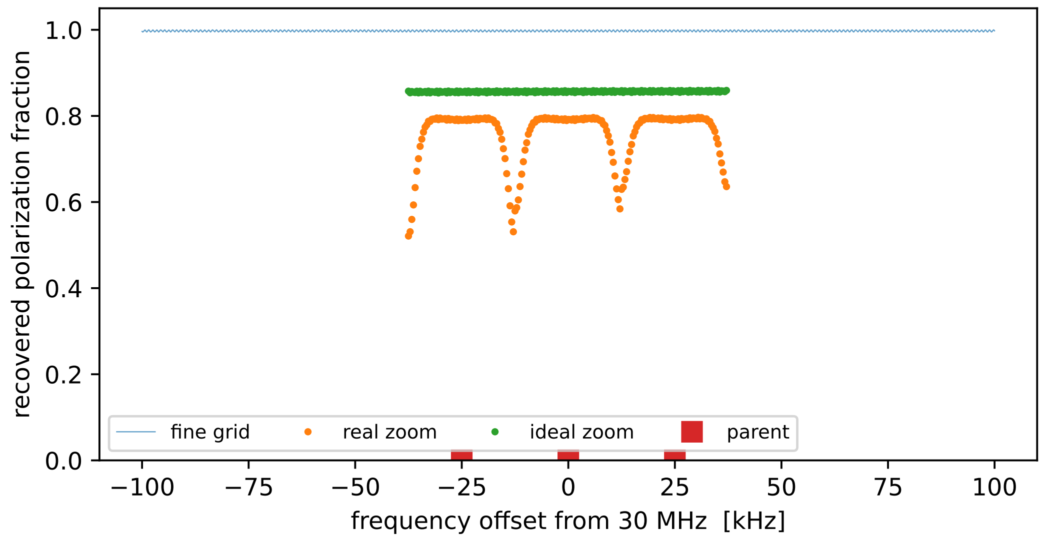

Figure 5: Recovered polarization fraction at transit per frequency bin. The fine grid sits at $p \simeq 1$ (rank-1 source); with the leakage nulled, the parent channels now show the true bandwidth depolarization, $p \simeq 7 \times 10^{-4}$; ideal zoom bins recover a flat $p \simeq 0.86$, while the real zoom bins (median 0.79) show characteristic dips at the parent-bin edges and centers caused by their aliased response tails.

Table 1: Fractional effect of sky polarization on the binned products, relative to the perfect-depolarization (I-only) run: time-median (max) of $|\Delta X|/I_{\rm obs}^{I\,only}$ over one sidereal day.

	central parent bin		central zoom bin	
band	ΔI	$\lvert\Delta P\rvert$	ΔI	$\lvert\Delta P\rvert$
10 MHz	$7.2\,(18) \times 10^{-6}$	$0.7\,(1.1) \times 10^{-4}$	$3.8\,(11) \times 10^{-5}$	$2.7\,(5.6) \times 10^{-4}$
30 MHz	$0.9\,(1.9) \times 10^{-4}$	$1.6\,(5.3) \times 10^{-4}$	$1.9\,(7.1) \times 10^{-4}$	$4.5\,(9.0) \times 10^{-4}$
50 MHz	$2.3\,(7.9) \times 10^{-4}$	$2.3\,(9.1) \times 10^{-4}$	$3.1\,(8.2) \times 10^{-4}$	$3.8\,(10) \times 10^{-4}$

7 Step 4: delay-space signature

Because the beam is frozen in frequency, the Faraday screen is the *only* source of spectral structure, and delay space separates it cleanly from the (chromatically smooth) sky: a screen of depth $\phi_{\rm FD}$ maps to the delay

$$\tau_{\rm FD}(\phi_{\rm FD}, \nu) = \frac{1}{2\pi}\left|\frac{d}{d\nu} 2\phi_{\rm FD}\lambda^2\right| = \frac{2\phi_{\rm FD}c^2}{\pi\nu^3} \approx 64\,\mu{\rm s}\left(\frac{\phi_{\rm FD}}{30\,{\rm rad\,m^{-2}}}\right)\left(\frac{\nu}{30\,{\rm MHz}}\right)^{-3}, \tag{3}$$

so the ensemble of rotation measures on the sky paints a delay "spectrum" that is a beam-weighted image of the RM distribution, stretched by the steep ν^{-3} lever arm.

Deconvolving the zoom channelization. The zoom stage runs a 64-point FFT on the critically sampled 25 kHz parent stream, so each bin has image responses folded about the parent-band edges: the Nyquist bin ($k = 32$) responds with an exact 50/50 split to the two boundary frequencies ±12.5 kHz of its parent (i.e. the slots halfway to each neighboring parent), and the near-boundary bins retain ∼ 40% images. Because these response shapes are known, the mapping from a spectrum gridded on the 192 nominal 390.625 Hz slots (padded by 32 slots per side to represent folds

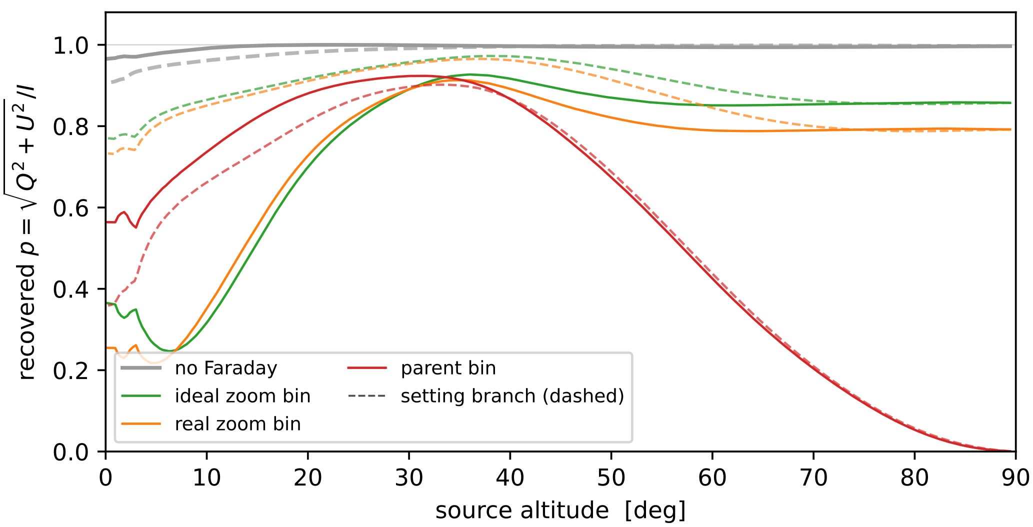

Figure 6: Recovered polarization fraction as a function of source altitude (solid: rising branch, dashed: setting branch; the two differ because the parallactic angle, and hence the instrument-frame EVPA, is not symmetric about transit). The fully polarized source has a rank-1 covariance, so $p \leq 1$ always, with the Faraday-free curve (grey) saturating the bound at essentially all altitudes. The 25 kHz parent bins (red) depolarize essentially completely near transit ($p \to 7 \times 10^{-4}$), where the intrinsic ∼1.9 kHz Faraday oscillation dominates the bin average; the residual floor equals the calibrated leakage of Fig. 2, since a fully depolarized bin leaves exactly the unpolarized-source covariance. Towards the horizon the parent bins recover more polarization while the zoom bins lose some, as beam-phase gradients across the bin become the limiting factor.

from just outside the band) to the 192 measured bins is an explicit linear transfer matrix — and a well-conditioned one (condition number 3.9). We invert it per time sample with a second-difference (smoothness) regularization, which is needed only to resolve the degeneracy of the parent-boundary and pad slots that enter solely through the shared Nyquist folds; a plain min-norm inverse leaves those few slots free and their errors ring across the whole delay range. The result is the "deconvolved" curve in Fig. 15.

Three practical conclusions follow. (i) The parent 25 kHz channels alone cannot reach the Faraday plateau at all: their Nyquist delay of 20 μs lies below $\tau_{\rm FD}$ of essentially every line of sight at $\nu \lesssim 30$ MHz. (ii) The 64-fold zoom sampling opens the window 50 μs $\lesssim |\tau| \lesssim 1.28$ ms, which at 30 MHz covers rotation measures from ~ 25 to ~ 600 rad m^{-2} and at 10 MHz resolves the ionospheric regime $\phi_{\rm FD} \lesssim 20$ rad m^{-2}. (iii) The measured zoom response aliases $\sim 10^{-3}$ of the strong low-delay power across the band, but because the bin transfer is known and well conditioned, a simple regularized linear inversion removes the aliasing floor in post-processing and restores the ideal-binning performance — at 50 MHz this is the difference between missing and cleanly resolving the $\tau_{\rm FD}(\phi_{\rm max})$ cutoff.

8 Conclusions

The four-port LuSEE-Night correlator, propagated through the real spectrometer channelization, separates Faraday rotation from the smooth foreground in two complementary regimes. A bright compact polarized source (Sec. 3) produces coherent polarization oscillations that the 25 kHz parent

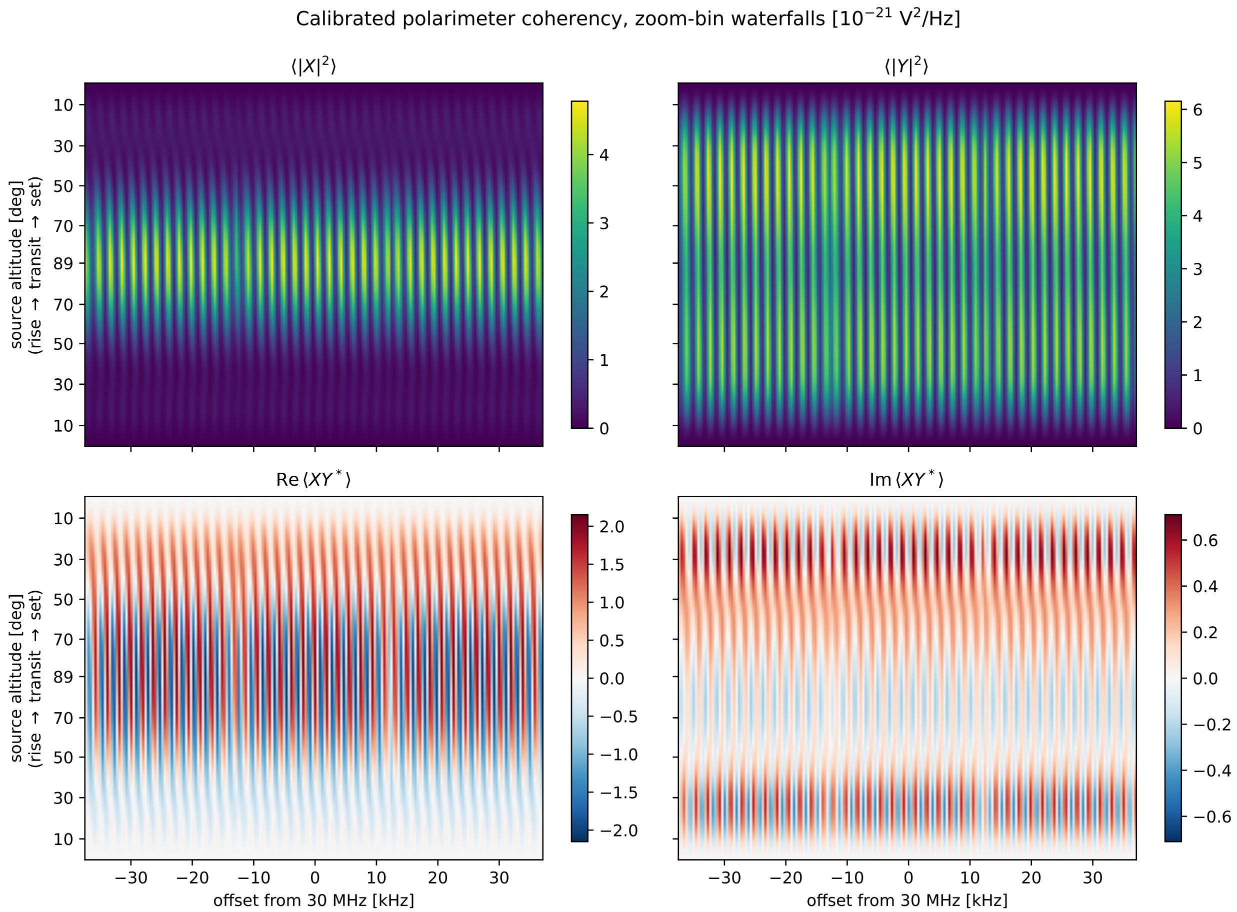

Figure 7: Coherency of the zenith-calibrated polarimeter in the zoom-bin channelization, against source altitude along the track (rise → transit → set) vs. zoom frequency: $\langle|X|^2\rangle$, $\langle|Y|^2\rangle$ (top) and $\mathrm{Re}\,\langle XY^*\rangle$, $\mathrm{Im}\,\langle XY^*\rangle$ (bottom), where X and Y are the calibrated four-port combinations of Sec. 2. The 1.9 kHz Faraday oscillation appears as vertical striping in all four panels; the slow structure along the track is the parallactic rotation of the source EVPA (which moves power between the X and Y autos) modulated by the altitude dependence of the beam.

channels erase (recovered $p \simeq 7 \times 10^{-4}$ with the leakage-nulled polarimeter, for $\phi_{\mathrm{FD}} = 250\ \mathrm{rad\,m^{-2}}$ at 30 MHz) but the zoom sub-bins recover at 79–86% amplitude. For the diffuse sky (Secs. 4–7) the many-RM mixture within the beam suppresses any single oscillation, yet the delay-space power spectrum of $Q_{\mathrm{obs}} + iU_{\mathrm{obs}}$ isolates the Faraday signal completely: with an achromatic beam all power at $\tau \neq 0$ is Faraday-induced, its envelope images the sky RM distribution through $\tau = 2\phi_{\mathrm{FD}}c^2/\pi\nu^3$, and its cutoff tracks the ν^{-3} scaling across the 10–50 MHz bands — a signature nothing else in the system mimics. The realistic zoom-bin response supports this measurement over two decades of delay with an aliasing floor $\sim 10^{-3}$ of the peak power.

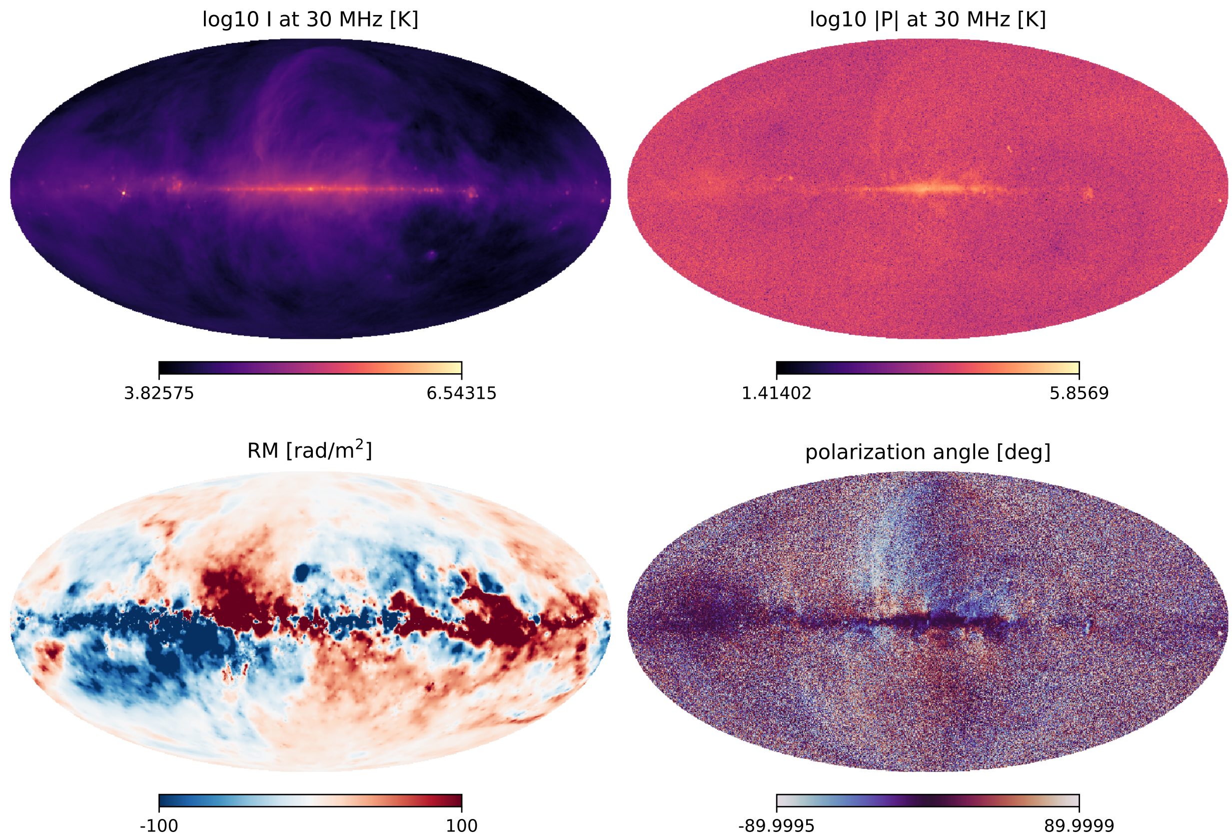

Figure 8: Input sky at 30 MHz: intensity, polarized intensity, Faraday depth and polarization angle (native $N_{\mathrm{side}} = 512$).

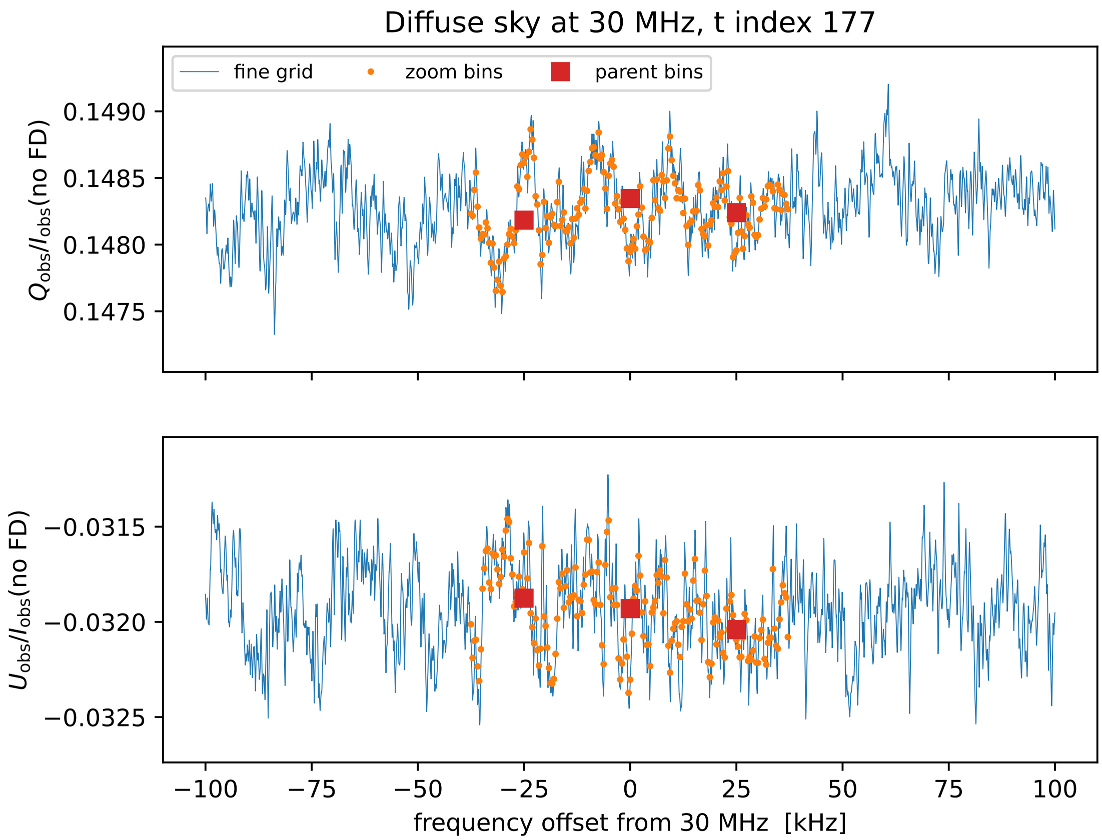

Figure 9: Beam-integrated $Q_{\rm obs}/I_{\rm obs}$, $U_{\rm obs}/I_{\rm obs}$ across the $\pm 100\,$kHz window at the time of maximum sky signal, with the y-range zoomed onto the fine-grid structure (the Faraday-free level lies far off scale). The smooth mean level is almost entirely instrumental leakage of the *unpolarized* emission (Sec. 6); the fast "hash" riding on it, at the few$\times 10^{-3}$ relative level, is the surviving sky polarization scrambled by the many rotation measures within the beam. The zoom bins trace a smoothed version of the hash while the parent bins average over it.

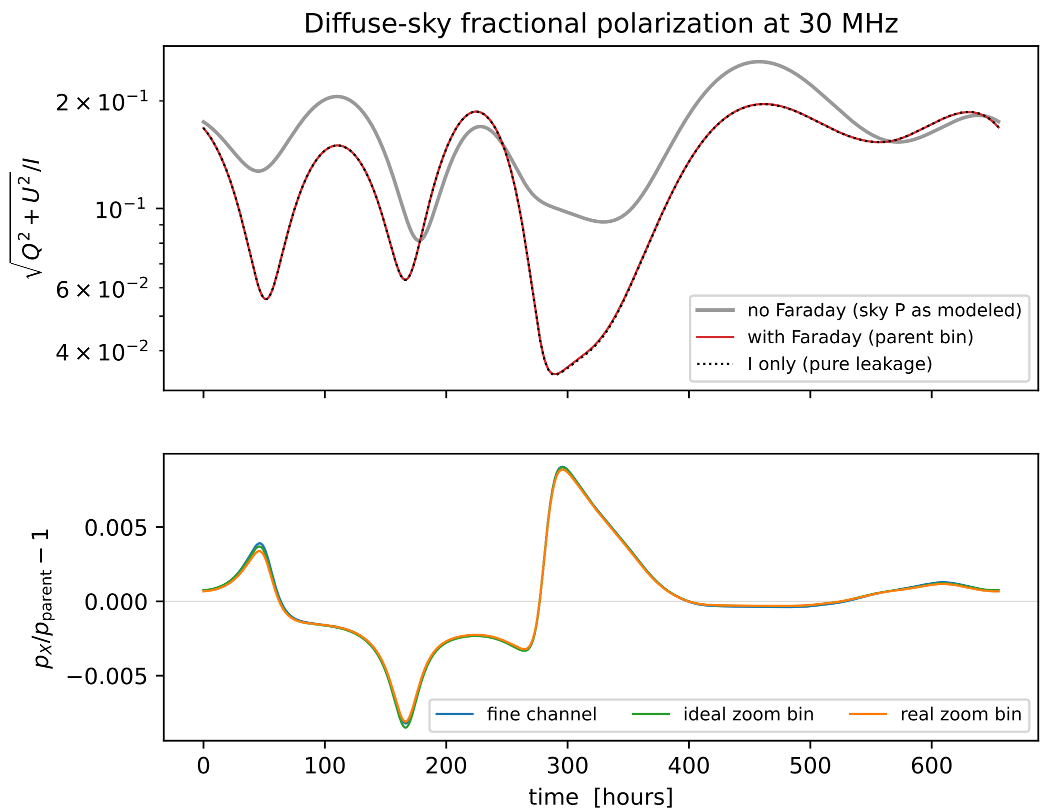

Figure 10: Fractional polarization of the beam-integrated signal over one sidereal day. Top: "no Faraday" keeps the sky polarization exactly as modeled (WMAP-scaled Q, U) with the Faraday screen switched off; "with Faraday" is the full simulation in the central parent bin; the dotted line is the $Q = U = 0$ (I-only) run, i.e. pure instrumental leakage. The Faraday curve lies exactly on the leakage curve — after beam-scale rotation-measure scrambling, the recovered polarization *is* the leakage track (cf. Sec. 6). Bottom: fine-channel, ideal-zoom and real-zoom values relative to the parent bin — all channelizations agree to better than 1%: at 30 MHz the diffuse-sky spectral structure is resolved even by the 25 kHz parent bins.

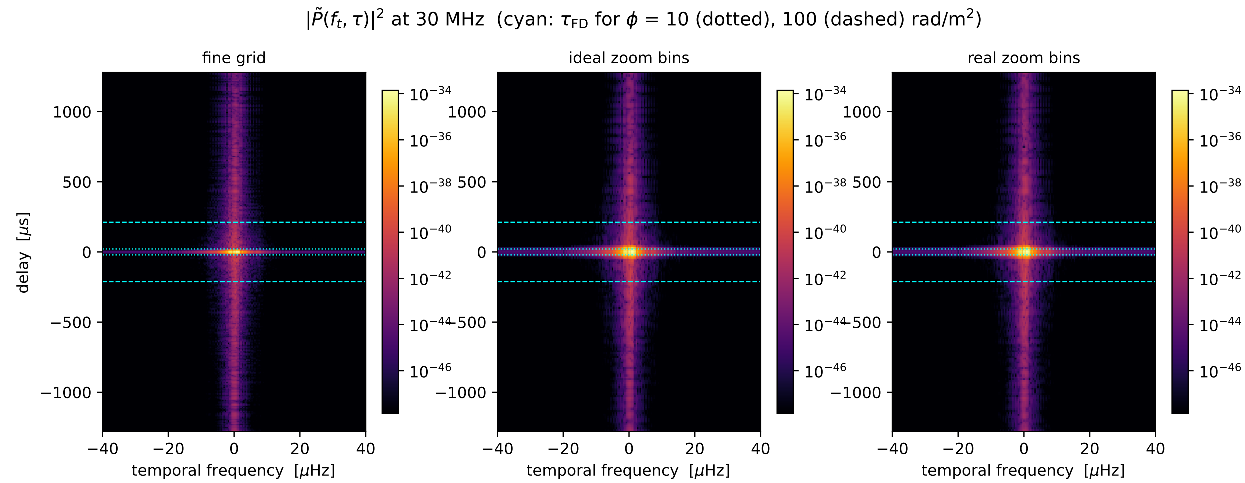

Figure 11: 2D power spectrum of $\tilde{P}_{\rm obs} = Q_{\rm obs} + iU_{\rm obs}$ over (temporal frequency, delay) at 30 MHz. With the beam frozen in frequency, *all* power at $\tau \neq 0$ is Faraday-induced. The cyan lines mark $\tau_{\rm FD} = 2\phi c^2/(\pi\nu^3)$ for $\phi = 10, 100$ rad m^{-2}. The real zoom bins (right) scatter noticeably more power to high delays than the ideal Gaussian bins (middle) — the aliasing artifact floor of the zoom channelization.

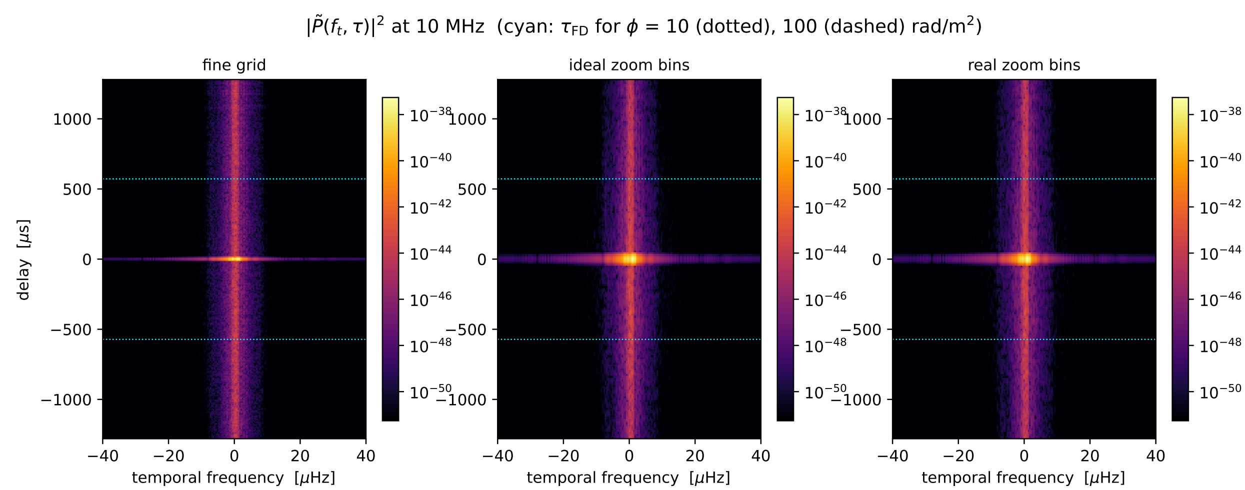

Figure 12: As Fig. 11, at 10 MHz. The Faraday tail now fills the entire delay range accessible to the zoom sampling ($|\tau| < 1.28$ ms): at 10 MHz one rad m^{-2} of Faraday depth corresponds to 57 μs of delay.

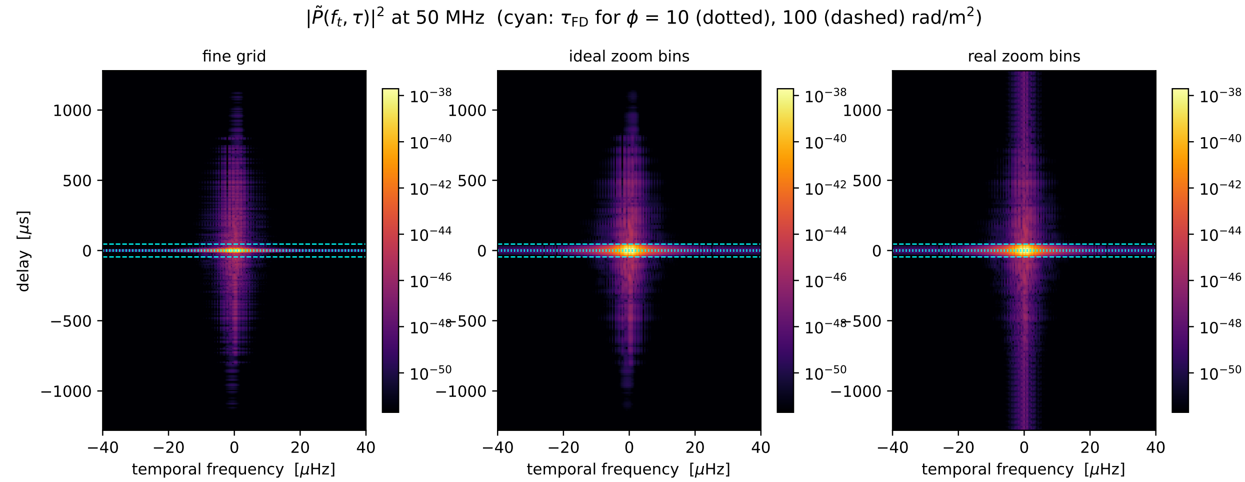

Figure 13: As Fig. 11, at 50 MHz. The Faraday signal is confined to a compact "diamond" with a sharp edge at $|\tau| \simeq 1.1\,\mathrm{ms}$ — the delay of the largest rotation measures on the sky ($|\phi| \simeq 2400\,\mathrm{rad\,m^{-2}}$ in the inner galactic plane). The real zoom bins (right panel) scatter a vertical stripe of aliased power far beyond the true envelope, which the ideal Gaussian bins do not.

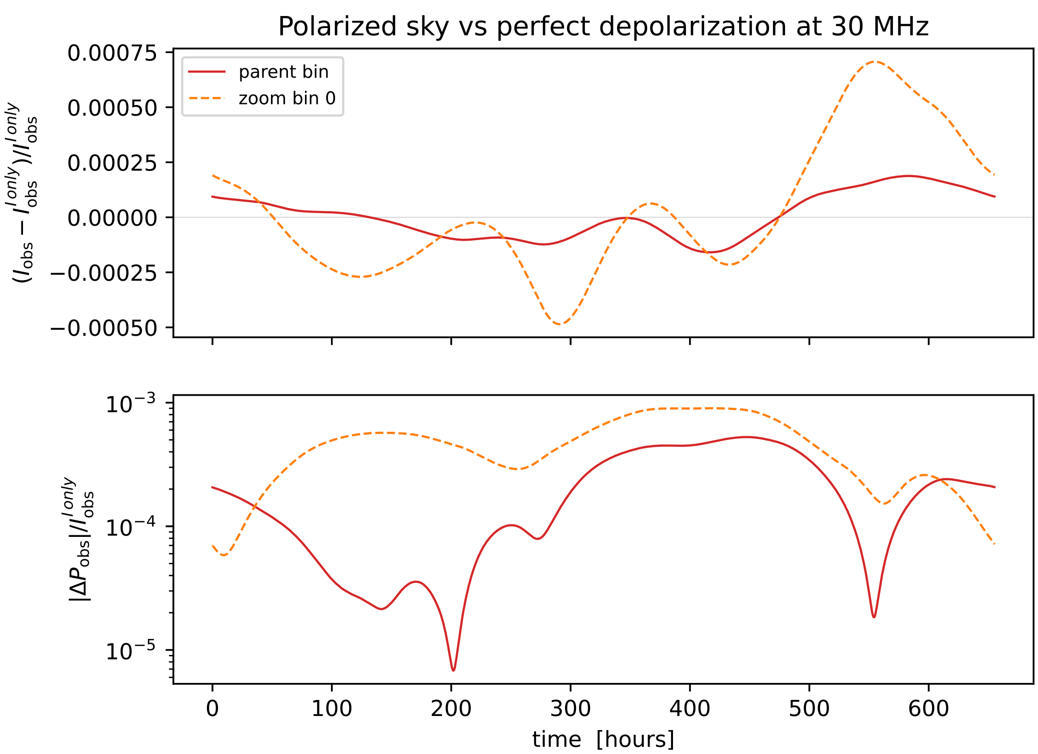

Figure 14: Fractional effect of sky polarization at 30 MHz over one sidereal day: intensity shift (top) and polarization shift (bottom) of the full-sky run relative to the I-only run, for the central parent bin and the central zoom bin. The zoom bin consistently retains a factor of a few more of the Faraday-surviving polarization.

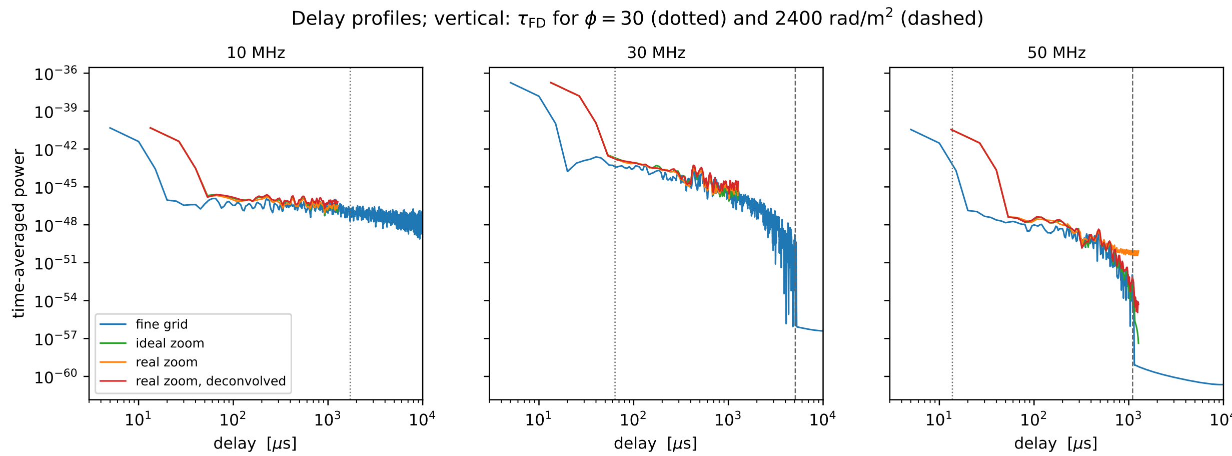

Figure 15: Time-averaged delay-power profiles of $\tilde{P}_{\rm obs}$ per band. The Faraday plateau extends from $\tau_{\rm FD}(\phi \sim \text{few})$ to a sharp cutoff at $\tau_{\rm FD}(\phi_{\max} \simeq 2400\ \text{rad}\,\text{m}^{-2})$ (dashed), clearly visible at 30 MHz ($\simeq 5$ ms) and 50 MHz ($\simeq 1.1$ ms) and beyond the plotted range at 10 MHz. The zoom bins reproduce the fine-grid profile faithfully between $\sim 50\,\mu$s (their leakage-broadened main lobe) and their Nyquist delay of 1.28 ms. At 50 MHz the raw *real* zoom bins (orange) flatten onto an aliasing floor about three decades below the plateau and miss the cutoff, while the ideal Gaussian bins keep following the true profile. The linear deconvolution of the known bin transfer (red) removes this floor: it tracks the fine-grid profile through the full cutoff, recovering ~ 4 decades below the raw aliasing floor with the real channelizer response.